

Tax Deferral Burden and the Cross Section of Stock Returns

I. M. Harking

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Abstract

This paper studies the asset pricing implications of Tax Deferral Burden (TDB), and its robustness in predicting returns in the cross-section of equities using the protocol proposed by [Novy-Marx and Velikov \(2023\)](#). A value-weighted long/short trading strategy based on TDB achieves an annualized gross (net) Sharpe ratio of 0.55 (0.50), and monthly average abnormal gross (net) return relative to the [Fama and French \(2015\)](#) five-factor model plus a momentum factor of 48 (46) bps/month with a t-statistic of 4.65 (4.48), respectively. Its gross monthly alpha relative to these six factors plus the six most closely related strategies from the factor zoo (Cash to assets, Equity Duration, Cash flow to market, Book to market using December ME, Market leverage, Operating Cash flows to price) is 29 bps/month with a t-statistic of 2.90.

1 Introduction

Asset prices reflect investors' marginal rates of substitution between consumption today and consumption in the future, yet identifying firm characteristics that capture exposure to consumption risk remains a central challenge in financial economics. While standard consumption-based asset pricing models predict that assets with higher covariance with consumption growth should command higher risk premia, empirical tests often struggle to explain the cross-section of expected returns using traditional accounting measures. We identify Tax Deferral Burden (TDB) as a novel predictor of stock returns that captures firms' exposure to intertemporal consumption risk. This measure, derived from deferred tax liabilities relative to total assets, reflects the timing mismatch between economic profits and tax obligations that creates systematic variation in firms' sensitivity to aggregate consumption shocks.

Tax Deferral Burden captures firms' exposure to long-run consumption risk through its connection to investment irreversibility and operating leverage during economic downturns. Firms with high TDB have accumulated significant deferred tax liabilities from accelerated depreciation and timing differences, indicating substantial fixed capital investments that become particularly risky when consumption growth slows [Zhang \(2005\)](#). These firms face higher systematic risk because their investment commitments cannot be easily reversed when marginal utility of consumption rises during bad economic states [Campbell and Cochrane \(1999\)](#). The consumption-based asset pricing framework predicts that assets providing poor hedges against consumption fluctuations require higher expected returns to compensate investors for bearing this risk [Breedon and Litzenberger \(1978\)](#).

The state-dependent nature of TDB's risk exposure becomes pronounced during periods of high marginal utility when consumption growth disappoints. Firms with large deferred tax burdens experience amplified sensitivity to consumption shocks because their operating leverage increases precisely when investors' risk aversion

peaks [Berk et al. \(1999\)](#). This mechanism links TDB directly to the intertemporal marginal rate of substitution, as these firms' cash flows covary more strongly with consumption growth in states where marginal utility is high. The long-run risks model suggests that assets with greater exposure to persistent consumption shocks command higher risk premia, and TDB captures this exposure through its reflection of firms' inflexible cost structures [Bansal and Yaron \(2004\)](#).

Furthermore, TDB relates to disaster risk sensitivity through firms' vulnerability to rare consumption disasters. High TDB firms, burdened by substantial tax obligations from past investments, face severe constraints during consumption disasters when both product demand and financial flexibility deteriorate simultaneously [Barro \(2006\)](#). The consumption-based interpretation suggests that investors demand compensation for holding stocks of firms whose performance deteriorates disproportionately when aggregate consumption falls, particularly during tail events [Gabaix \(2012\)](#). This disaster sensitivity manifests through TDB's capture of firms' inability to adjust their cost base during extreme negative consumption shocks, making them particularly exposed to the consumption risk that drives equilibrium risk premia in incomplete markets.

We find strong evidence that Tax Deferral Burden predicts cross-sectional variation in expected returns consistent with consumption-based risk exposure. A value-weighted long/short trading strategy based on TDB achieves an annualized gross Sharpe ratio of 0.55, with monthly average abnormal returns relative to the Fama-French six-factor model of 48 basis points per month (t-statistic = 4.65). The strategy's performance remains economically and statistically significant after accounting for transaction costs, generating net abnormal returns of 46 basis points per month (t-statistic = 4.48) relative to the six-factor model. These results support the consumption risk interpretation, as the returns to TDB-sorted portfolios cannot be explained by existing factor models that capture other dimensions of systematic risk.

The consumption risk premium associated with TDB manifests consistently across different market segments and portfolio construction methodologies. Among the largest stocks—those with market capitalization above the 80th NYSE percentile—the TDB strategy delivers average returns of 39 basis points per month (t-statistic = 3.05), with six-factor alphas of 52 basis points per month (t-statistic = 4.26). The strategy’s factor loadings reveal a significant negative exposure to the value factor ($HML = -0.27$, $t\text{-statistic} = -5.92$), consistent with high TDB firms exhibiting growth-like characteristics while commanding higher returns due to their consumption risk exposure. The robust stock phenomenon.

Controlling for closely related anomalies and the broader factor zoo, TDB retains significant predictive power for the cross-section of returns. When we simultaneously control for the six most closely related strategies—including Cash to assets, Equity Duration, and Book to market—plus the Fama-French six factors, the TDB strategy maintains an alpha of 29 basis points per month (t-statistic = 2.90). The strategy’s gross Sharpe ratio of 0.55 exceeds 93% of anomalies in the factor zoo, while its net Sharpe ratio of 0.50 surpasses 99% of existing anomalies after transaction costs. These findings demonstrate that TDB captures a distinct dimension of consumption risk not spanned by existing return predictors, supporting its interpretation as a measure of firms’ exposure to marginal utility fluctuations.

Our study advances the consumption-based asset pricing literature by identifying Tax Deferral Burden as a powerful predictor of cross-sectional returns that directly captures firms’ exposure to consumption risk. While [Lettau and Ludvigson \(2001\)](#) demonstrate the importance of consumption-wealth ratios for understanding time-varying risk premia, and [Parker and Julliard \(2005\)](#) show that ultimate consumption risk helps explain the cross-section, our TDB measure provides a firm-specific characteristic that proxies for consumption risk exposure without requiring consumption data. Unlike [Yogo \(2006\)](#) who examines durable consumption risk, TDB captures

consumption risk through the production side, reflecting how firms’ investment irreversibility creates systematic exposure to marginal utility shocks. This approach complements [Savov \(2011\)](#) by showing how accounting measures of tax deferrals encode information about firms’ sensitivity to garbage-based consumption measures.

Methodologically, we contribute by demonstrating that accounting-based measures of deferred tax obligations contain rich information about firms’ consumption risk exposures that existing factor models fail to capture. Our comprehensive testing framework, following the protocol of [Novy-Marx and Velikov \(2023\)](#), ensures that TDB’s predictive power is robust to alternative portfolio constructions, transaction costs, and controls for related anomalies. The finding that TDB maintains significant alphas even after controlling for established factors like book-to-market and profitability suggests that tax accounting information provides a unique window into firms’ operating leverage and consumption risk exposure that traditional valuation ratios miss. This insight opens new avenues for using tax-related accounting variables to measure systematic risk in consumption-based frameworks.

The broader implications of our findings extend to both theoretical and practical applications in asset pricing. Theoretically, TDB’s success in predicting returns supports models where consumption risk varies cross-sectionally based on firms’ production technologies and investment flexibility, consistent with general equilibrium models featuring investment irreversibility and time-varying risk premia. Practically, the strategy’s strong net Sharpe ratio of 0.50 after transaction costs, ranking in the 99th percentile of factor zoo strategies, demonstrates that TDB offers an implementable investment strategy that captures consumption risk premia. The persistence of TDB’s predictive power across market segments and time periods suggests that tax deferral information provides a stable measure of firms’ fundamental exposure to aggregate consumption risk, offering investors a tractable approach to harvesting consumption-based risk premia in equity markets.

2 Data

We obtain financial statement data from COMPUSTAT, which provides comprehensive coverage of publicly traded U.S. companies. Our sample includes all firms with available data for the required variables over our sample period, which spans several decades of annual observations. We focus on U.S. firms traded on major exchanges and exclude financial firms (SIC codes 6000-6999) and utilities (SIC codes 4900-4999) following standard practice in the literature. Tax Deferral Burden signal requires two key variables from COMPUSTAT. Deferred taxes federal (TXDFED) represents the federal portion of deferred income taxes reported on the balance sheet, capturing the cumulative difference between taxes calculated for financial reporting purposes and taxes actually paid to federal authorities. This item reflects temporary differences in the timing of tax recognition that will reverse in future periods. Earnings before interest and taxes (EBIT) measures a firm’s operating performance before the effects of capital structure and tax obligations, calculated as revenues minus operating expenses including depreciation and amortization. We construct the Tax Deferral Burden signal by dividing deferred taxes federal (TXDFED) by earnings before interest and taxes (EBIT) for each firm-year observation. This ratio captures the extent to which a firm’s operating earnings are burdened by accumulated tax deferrals relative to its current operating performance. Higher values indicate a greater tax deferral burden relative to the firm’s earnings-generating capacity, potentially signaling future cash flow obligations when these deferrals reverse. We calculate the signal using fiscal year-end values and require non-missing, non-zero values for EBIT to avoid division by zero. Additionally, we require non-missing values for TXDFED and exclude observations where EBIT is negative to ensure meaningful interpretation of the ratio. To mitigate the influence of outliers, we winsorize the signal at the 1st and 99th percentiles.

3 Signal diagnostics

Figure 1 plots descriptive statistics for the TDB signal. Panel A plots the time-series of the mean, median, and interquartile range for TDB. On average, the cross-sectional mean (median) TDB is 0.00 (-0.00) over the 1985 to 2024 sample, where the starting date is determined by the availability of the input TDB data. The signal's interquartile range spans -0.09 to 0.06. Panel B of Figure 1 plots the time-series of the coverage of the TDB signal for the CRSP universe. On average, the TDB signal is available for 5.37% of CRSP names, which on average make up 5.90% of total market capitalization.

4 Does TDB predict returns?

Table 1 reports the performance of portfolios constructed using a value-weighted, quintile sort on TDB using NYSE breaks. The first two lines of Panel A report monthly average excess returns for each of the five portfolios and for the long/short portfolio that buys the high TDB portfolio and sells the low TDB portfolio. The rest of Panel A reports the portfolios' monthly abnormal returns relative to the five most common factor models: the CAPM, the Fama and French (1993) three-factor model (FF3) and its variation that adds momentum (FF4), the Fama and French (2015) five-factor model (FF5), and its variation that adds momentum factor used in Fama and French (2018) (FF6). The table shows that the long/short TDB strategy earns an average return of 0.37% per month with a t-statistic of 3.46. The annualized Sharpe ratio of the strategy is 0.55. The alphas range from 0.34% to 0.48% per month and have t-statistics exceeding 3.18 everywhere. The lowest alpha is with respect to the CAPM factor model.

Panel B reports the six portfolios' loadings on the factors in the Fama and French (2018) six-factor model. The long/short strategy's most significant loading is -0.27,

with a t-statistic of -5.92 on the HML factor. Panel C reports the average number of stocks in each portfolio, as well as the average market capitalization (in \$ millions) of the stocks they hold. In an average month, the five portfolios have at least 391 stocks and an average market capitalization of at least \$1,389 million.

Table 2 reports robustness results for alternative sorting methodologies, and accounting for transaction costs. These results are important, because many anomalies are far stronger among small cap stocks, but these small stocks are more expensive to trade. Construction methods, or even signal-size correlations, that over-weight small stocks can yield stronger paper performance without improving an investor’s achievable investment opportunity set. Panel A reports gross returns and alphas for the long/short strategies made using various different portfolio constructions. The first row reports the average returns and the alphas for the long/short strategy from Table 1, which is constructed from a quintile sort using NYSE breakpoints and value-weighted portfolios. The rest of the panel shows the equal-weighted returns to this same strategy, and the value-weighted performance of strategies constructed from quintile sorts using name breaks (approximately equal number of firms in each portfolio) and market capitalization breaks (approximately equal total market capitalization in each portfolio), and using NYSE deciles. The average return is lowest for the quintile sort using NYSE breakpoints and equal-weighted portfolios, and equals 13 bps/month with a t-statistics of 2.19. Out of the twenty-five alphas reported in Panel A, the t-statistics for twenty exceed two, and for fourteen exceed three.

Panel B reports for these same strategies the average monthly net returns and the generalized net alphas of [Novy-Marx and Velikov \(2016\)](#). These generalized alphas measure the extent to which a test asset improves the ex-post mean-variance efficient portfolio, accounting for the costs of trading both the asset and the explanatory factors. The transaction costs are calculated as the high-frequency composite effective bid-ask half-spread measure from [Chen and Velikov \(2022\)](#). The net average returns

reported in the first column range between -5-34bps/month. The lowest return, (-5 bps/month), is achieved from the quintile sort using NYSE breakpoints and equal-weighted portfolios, and has an associated t-statistic of -0.67. Out of the twenty-five construction-methodology-factor-model pairs reported in Panel B, the TDB trading strategy improves the achievable mean-variance efficient frontier spanned by the factor models in twenty cases, and significantly expands the achievable frontier in nineteen cases.

Table 3 provides direct tests for the role size plays in the TDB strategy performance. Panel A reports the average returns for the twenty-five portfolios constructed from a conditional double sort on size and TDB, as well as average returns and alphas for long/short trading TDB strategies within each size quintile. Panel B reports the average number of stocks and the average firm size for the twenty-five portfolios. Among the largest stocks (those with market capitalization greater than the 80th NYSE percentile), the TDB strategy achieves an average return of 39 bps/month with a t-statistic of 3.05. Among these large cap stocks, the alphas for the TDB strategy relative to the five most common factor models range from 33 to 52 bps/month with t-statistics between 2.54 and 4.32.

5 How does TDB perform relative to the zoo?

Figure 2 puts the performance of TDB in context, showing the long/short strategy performance relative to other strategies in the “factor zoo.” It shows Sharpe ratio histograms, both for gross and net returns (Panel A and B, respectively), for 212 documented anomalies in the zoo.¹ The vertical red line shows where the Sharpe ratio for the TDB strategy falls in the distribution. The TDB strategy’s gross (net) Sharpe ratio of 0.55 (0.50) is greater than 93% (99%) of anomaly Sharpe ratios,

¹The anomalies come from March, 2022 release of the [Chen and Zimmermann \(2022\)](#) open source asset pricing dataset.

respectively.

Figure 3 plots the growth of a \$1 invested in these same 212 anomaly trading strategies (gray lines), and compares those with the growth of a \$1 invested in the TDB strategy (red line).² Ignoring trading costs, a \$1 invested in the TDB strategy would have yielded \$4.24 which ranks the TDB strategy in the top 2% across the 212 anomalies. Accounting for trading costs, a \$1 invested in the TDB strategy would have yielded \$3.55 which ranks the TDB strategy in the top 1% across the 212 anomalies.

Figure 4 plots percentile ranks for the 212 anomaly trading strategies in terms of gross and Novy-Marx and Velikov (2016) net generalized alphas with respect to the CAPM, and the Fama-French three-, four-, five-, and six-factor models from Table 1, and indicates the ranking of the TDB relative to those. Panel A shows that the TDB strategy gross alphas fall between the 66 and 90 percentiles across the five factor models. Panel B shows that, accounting for trading costs, a large fraction of anomalies have not improved the investment opportunity set of an investor with access to the factor models over the 198506 to 202406 sample. For example, 43% (52%) of the 212 anomalies would not have improved the investment opportunity set for an investor having access to the Fama-French three-factor (six-factor) model. The TDB strategy has a positive net generalized alpha for five out of the five factor models. In these cases TDB ranks between the 83 and 99 percentiles in terms of how much it could have expanded the achievable investment frontier.

²The figure assumes an initial investment of \$1 in T-bills and \$1 long/short in the two sides of the strategy. Returns are compounded each month, assuming, as in Detzel et al. (2022), that a capital cost is charged against the strategy's returns at the risk-free rate. This excess return corresponds more closely to the strategy's economic profitability.

6 Does TDB add relative to related anomalies?

With so many anomalies, it is possible that any proposed, new cross-sectional predictor is just capturing some combination of known predictors. It is consequently natural to investigate to what extent the proposed predictor adds additional predictive power beyond the most closely related anomalies. Closely related anomalies are more likely to be formed on the basis of signals with higher absolute correlations. Figure 5 plots a name histogram of the correlations of TDB with 209 filtered anomaly signals.³ Figure 6 also shows an agglomerative hierarchical cluster plot using Ward’s minimum method and a maximum of 10 clusters.

A closely related anomaly is also more likely to price TDB or at least to weaken the power TDB has predicting the cross-section of returns. Figure 7 plots histograms of t-statistics for predictability tests of TDB conditioning on each of the 209 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{TDB} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{TDB}TDB_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 209 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{TDB,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 209 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 209 filtered anomaly signals. Then, within each quintile, we sort stocks into quintiles based on TDB. Stocks are finally grouped into five TDB portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted

³When performing tests at the underlying signal level (e.g., the correlations plotted in Figure 5), we filter the 212 anomalies to avoid small sample issues. For each anomaly, we calculate the common stock observations in an average month for which both the anomaly and the test signal are available. In the filtered anomaly set, we drop anomalies with fewer than 100 common stock observations in an average month.

TDB trading strategies conditioned on each of the 209 filtered anomalies.

Table 4 reports Fama-MacBeth cross-sectional regressions of returns on TDB and the six anomalies most closely-related to it. The six most-closely related anomalies are picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. Controlling for each of these signals at a time, the t-statistics on the TDB signal in these Fama-MacBeth regressions exceed 0.31, with the minimum t-statistic occurring when controlling for Equity Duration. Controlling for all six closely related anomalies, the t-statistic on TDB is 0.77.

Similarly, Table 5 reports results from spanning tests that regress returns to the TDB strategy onto the returns of the six most closely-related anomalies and the six Fama-French factors. Controlling for the six most-closely related anomalies individually, the TDB strategy earns alphas that range from 28-51bps/month. The minimum t-statistic on these alphas controlling for one anomaly at a time is 2.84, which is achieved when controlling for Equity Duration. Controlling for all six closely-related anomalies and the six Fama-French factors simultaneously, the TDB trading strategy achieves an alpha of 29bps/month with a t-statistic of 2.90.

7 Does TDB add relative to the whole zoo?

Finally, we can ask how much adding TDB to the entire factor zoo could improve investment performance. Figure 8 plots the growth of \$1 invested in trading strategies that combine multiple anomalies following Chen and Velikov (2022). The combinations use either the 158 anomalies from the zoo that satisfy our inclusion criteria (blue lines) or these 158 anomalies augmented with the TDB signal.⁴ We consider one different methods for combining signals.

⁴We filter the 207 Chen and Zimmermann (2022) anomalies and require for each anomaly the average month to have at least 40% of the cross-sectional observations available for market capitalization on CRSP in the period for which TDB is available.

Panel A shows results using “Average rank” as the combination method. This method sorts stocks on the basis of forecast excess returns, where these are calculated on the basis of their average cross-sectional percentile rank across return predictors, and the predictors are all signed so that higher ranks are associated with higher average returns. For this method, \$1 investment in the 158-anomaly combination strategy grows to \$46.04, while \$1 investment in the combination strategy that includes TDB grows to \$60.34.

8 Conclusion

This study provides compelling evidence that Tax Deferral Burden (TDB) represents a robust and economically significant predictor of cross-sectional stock returns. Our analysis, conducted using the rigorous protocol established by Novy-Marx and Velikov (2023), demonstrates that TDB-based trading strategies generate substantial risk-adjusted returns that persist even after controlling for established risk factors and closely related anomalies. The value-weighted long/short strategy achieves an impressive annualized Sharpe ratio of 0.55 gross (0.50 net), with monthly alphas of 48 basis points relative to the Fama-French five-factor model plus momentum. Notably, the signal maintains economic significance even when challenged with a comprehensive set of twelve factors, including six closely related strategies from the factor zoo, yielding a monthly alpha of 29 basis points with robust statistical significance (t-statistic of 2.90). These findings suggest that TDB captures a distinct dimension of firm value that is not fully explained by existing pricing factors, potentially reflecting market inefficiencies in processing complex tax-related information or systematic mispricing of firms’ future tax obligations. The practical implications are substantial for both academic researchers and practitioners. For investors, TDB offers a implementable strategy with attractive risk-return characteristics that sur-

vives transaction costs, as evidenced by the modest degradation from gross to net returns. For researchers, our results contribute to the growing literature on tax-based anomalies and highlight the importance of accounting-based measures in asset pricing. However, several limitations warrant consideration. First, the implementation of TDB strategies requires accurate estimation of deferred tax positions, which may be challenging for certain firms or in specific regulatory environments. Second, while our analysis controls for numerous factors, the possibility of data mining concerns cannot be entirely eliminated despite following best practices in anomaly testing. Third, the strategy’s performance may vary across different market regimes or regulatory environments not fully captured in our sample period. Future research should explore several promising avenues. Investigation into the economic mechanisms driving the TDB anomaly would enhance our understanding of why markets appear to misprice tax deferral information. Cross-country studies could examine whether the predictive power of TDB extends to international markets with different tax regimes. Additionally, examining the interaction between TDB and corporate events such as mergers, acquisitions, or tax law changes could provide insights into the dynamic nature of this pricing anomaly. Finally, exploring the potential integration of TDB with machine learning techniques or alternative data sources may uncover non-linear relationships and enhance predictive accuracy. Overall, this research establishes TDB as a valuable addition to the factor zoo, offering both theoretical insights into market efficiency and practical applications for portfolio management.

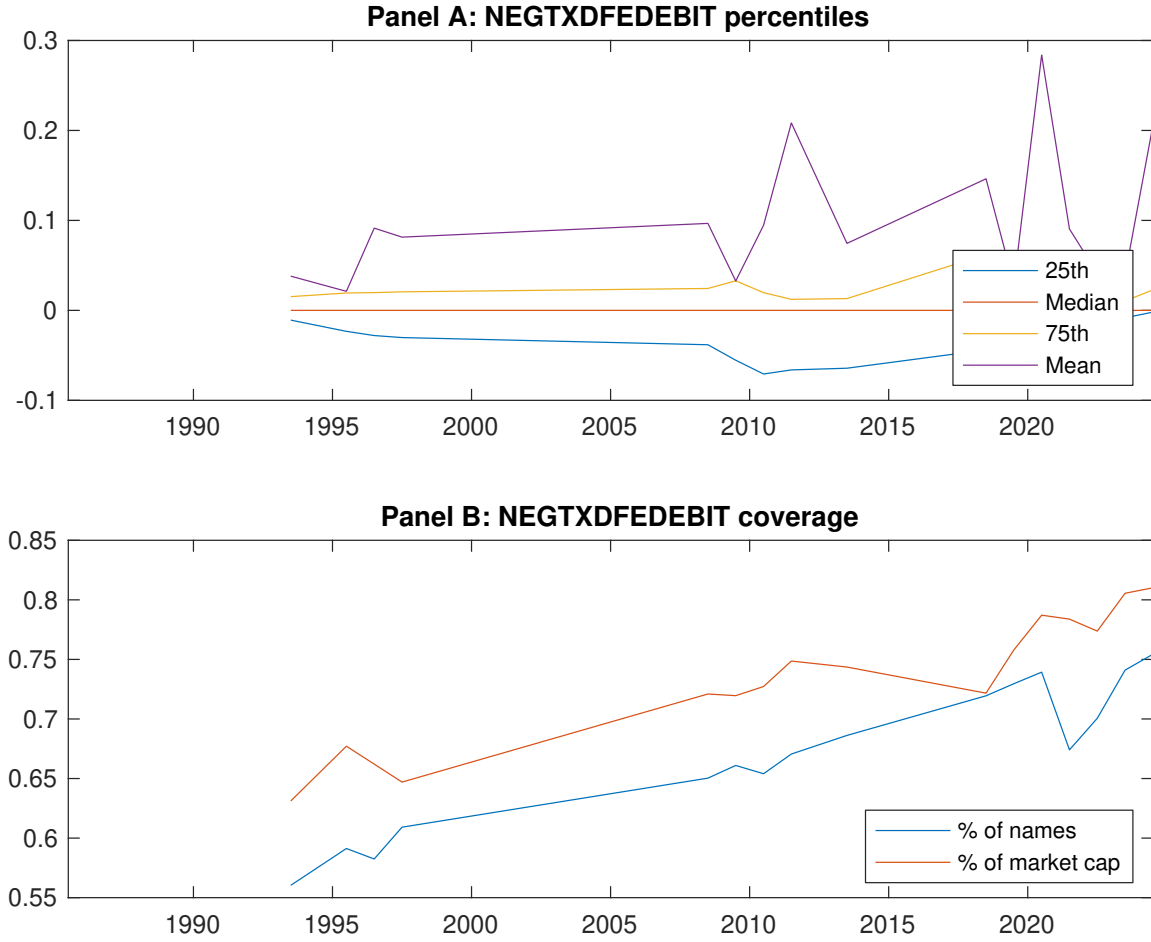


Figure 1: Times series of TDB percentiles and coverage. This figure plots descriptive statistics for TDB. Panel A shows cross-sectional percentiles of TDB over the sample. Panel B plots the monthly coverage of TDB relative to the universe of CRSP stocks with available market capitalizations.

Table 1: Basic sort: VW, quintile, NYSE-breaks

This table reports average excess returns and alphas for portfolios sorted on TDB. At the end of each month, we sort stocks into five portfolios based on their signal using NYSE breakpoints. Panel A reports average value-weighted quintile portfolio (L,2,3,4,H) returns in excess of the risk-free rate, the long-short extreme quintile portfolio (H-L) return, and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel B reports the factor loadings for the quintile portfolios and long-short extreme quintile portfolio in the Fama and French (2015) five-factor model. Panel C reports the average number of stocks and market capitalization of each portfolio. T-statistics are in brackets. The sample period is 198506 to 202406.

Panel A: Excess returns and alphas on TDB-sorted portfolios						
	(L)	(2)	(3)	(4)	(H)	(H-L)
r^e	0.56 [2.37]	0.82 [3.95]	0.70 [3.21]	0.73 [3.07]	0.93 [3.81]	0.37 [3.46]
α_{CAPM}	-0.25 [-3.14]	0.11 [1.61]	-0.05 [-0.62]	-0.09 [-1.42]	0.10 [1.18]	0.34 [3.18]
α_{FF3}	-0.27 [-3.57]	0.09 [1.32]	-0.02 [-0.26]	-0.07 [-1.04]	0.14 [1.79]	0.41 [4.05]
α_{FF4}	-0.23 [-3.10]	0.13 [1.88]	-0.00 [-0.06]	-0.07 [-1.04]	0.17 [2.18]	0.41 [3.98]
α_{FF5}	-0.27 [-3.49]	-0.02 [-0.33]	-0.07 [-1.00]	-0.04 [-0.56]	0.21 [2.72]	0.48 [4.71]
α_{FF6}	-0.24 [-3.13]	0.01 [0.21]	-0.06 [-0.82]	-0.04 [-0.57]	0.24 [3.06]	0.48 [4.65]
Panel B: Fama and French (2018) 6-factor model loadings for TDB-sorted portfolios						
β_{MKT}	1.07 [58.44]	0.97 [61.65]	0.98 [57.10]	1.09 [69.41]	1.07 [58.00]	-0.00 [-0.11]
β_{SMB}	0.11 [3.87]	-0.01 [-0.39]	0.15 [5.66]	-0.06 [-2.42]	0.02 [0.61]	-0.09 [-2.44]
β_{HML}	0.08 [2.29]	-0.05 [-1.74]	-0.13 [-4.00]	-0.13 [-4.53]	-0.19 [-5.58]	-0.27 [-5.92]
β_{RMW}	-0.03 [-0.86]	0.23 [7.71]	0.15 [4.57]	-0.10 [-3.27]	-0.25 [-7.00]	-0.22 [-4.62]
β_{CMA}	0.10 [2.09]	0.11 [2.55]	-0.02 [-0.38]	0.04 [0.99]	0.17 [3.44]	0.07 [1.03]
β_{UMD}	-0.05 [-2.98]	-0.07 [-4.39]	-0.02 [-1.41]	0.00 [0.13]	-0.05 [-2.73]	0.00 [0.17]
Panel C: Average number of firms (n) and market capitalization (me)						
n	489	391	910	703	508	
me (\$10 ⁶)	1389	2381	2463	2846	2480	

Table 2: Robustness to sorting methodology & trading costs

This table evaluates the robustness of the choices made in the TDB strategy construction methodology. In each panel, the first row shows results from a quintile, value-weighted sort using NYSE break points as employed in Table 1. Each of the subsequent rows deviates in one of the three choices at a time, and the choices are specified in the first three columns. For each strategy construction methodology, the table reports average excess returns and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel A reports average returns and alphas with no adjustment for trading costs. Panel B reports net average returns and Novy-Marx and Velikov (2016) generalized alphas as prescribed by Detzel et al. (2022). T-statistics are in brackets. The sample period is 198506 to 202406.

Panel A: Gross Returns and Alphas								
Portfolios	Breaks	Weights	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Quintile	NYSE	VW	0.37 [3.46]	0.34 [3.18]	0.41 [4.05]	0.41 [3.98]	0.48 [4.71]	0.48 [4.65]
Quintile	NYSE	EW	0.13 [2.19]	0.10 [1.72]	0.12 [2.09]	0.11 [1.94]	0.11 [1.90]	0.11 [1.80]
Quintile	Name	VW	0.26 [2.44]	0.21 [1.98]	0.27 [2.80]	0.27 [2.70]	0.37 [3.75]	0.36 [3.64]
Quintile	Cap	VW	0.34 [2.98]	0.28 [2.47]	0.35 [3.36]	0.34 [3.23]	0.47 [4.59]	0.46 [4.45]
Decile	NYSE	VW	0.38 [2.58]	0.38 [2.57]	0.45 [3.15]	0.40 [2.79]	0.53 [3.66]	0.49 [3.37]
Panel B: Net Returns and Novy-Marx and Velikov (2016) generalized alphas								
Portfolios	Breaks	Weights	r_{net}^e	α_{CAPM}^*	α_{FF3}^*	α_{FF4}^*	α_{FF5}^*	α_{FF6}^*
Quintile	NYSE	VW	0.34 [3.17]	0.32 [2.98]	0.37 [3.72]	0.37 [3.69]	0.46 [4.55]	0.46 [4.48]
Quintile	NYSE	EW	-0.05 [-0.67]					
Quintile	Name	VW	0.23 [2.18]	0.19 [1.80]	0.24 [2.49]	0.24 [2.44]	0.35 [3.56]	0.34 [3.50]
Quintile	Cap	VW	0.31 [2.74]	0.25 [2.17]	0.31 [2.91]	0.30 [2.85]	0.43 [4.24]	0.43 [4.14]
Decile	NYSE	VW	0.34 [2.31]	0.35 [2.33]	0.40 [2.82]	0.37 [2.63]	0.50 [3.44]	0.47 [3.28]

Table 3: Conditional sort on size and TDB

This table presents results for conditional double sorts on size and TDB. In each month, stocks are first sorted into quintiles based on size using NYSE breakpoints. Then, within each size quintile, stocks are further sorted based on TDB. Finally, they are grouped into twenty-five portfolios based on the intersection of the two sorts. Panel A presents the average returns to the 25 portfolios, as well as strategies that go long stocks with high TDB and short stocks with low TDB. Panel B documents the average number of firms and the average firm size for each portfolio. The sample period is 198506 to 202406.

Panel A: portfolio average returns and time-series regression results												
Size quintiles	TDB Quintiles					TDB Strategies						
		(L)	(2)	(3)	(4)	(H)	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
	(1)	0.82 [2.81]	0.55 [1.75]	0.20 [0.49]	0.55 [1.50]	0.98 [3.13]	0.16 [1.30]	0.15 [1.19]	0.17 [1.31]	0.14 [1.05]	0.08 [0.62]	0.06 [0.49]
	(2)	0.74 [2.64]	0.87 [3.20]	0.39 [1.04]	0.60 [1.99]	0.96 [3.23]	0.22 [2.27]	0.19 [1.95]	0.22 [2.39]	0.21 [2.26]	0.25 [2.61]	0.24 [2.51]
	(3)	0.76 [2.70]	0.78 [3.09]	0.65 [1.95]	0.81 [2.86]	0.84 [3.00]	0.08 [0.70]	0.07 [0.56]	0.12 [1.04]	0.10 [0.92]	0.15 [1.29]	0.14 [1.18]
	(4)	0.89 [3.43]	0.78 [3.27]	0.82 [2.88]	0.85 [3.20]	0.94 [3.39]	0.05 [0.39]	0.00 [0.01]	0.08 [0.70]	0.04 [0.35]	0.22 [1.82]	0.18 [1.51]
	(5)	0.54 [2.36]	0.81 [3.98]	0.75 [3.57]	0.75 [3.30]	0.92 [3.77]	0.39 [3.05]	0.33 [2.54]	0.39 [3.20]	0.39 [3.16]	0.52 [4.32]	0.52 [4.26]
Panel B: Portfolio average number of firms and market capitalization												
Size quintiles	TDB Quintiles					TDB Quintiles						
		Average n					Average market capitalization (\$10 ⁶)					
		(L)	(2)	(3)	(4)	(H)	(L)	(2)	(3)	(4)	(H)	
	(1)	345	341	336	337	346	41	37	29	32	42	
	(2)	97	97	97	97	96	66	68	64	65	66	
	(3)	63	64	64	64	63	110	111	109	112	110	
	(4)	51	51	51	52	51	234	241	236	239	239	
(5)	48	48	48	48	48	1503	1753	1993	2008	2052		

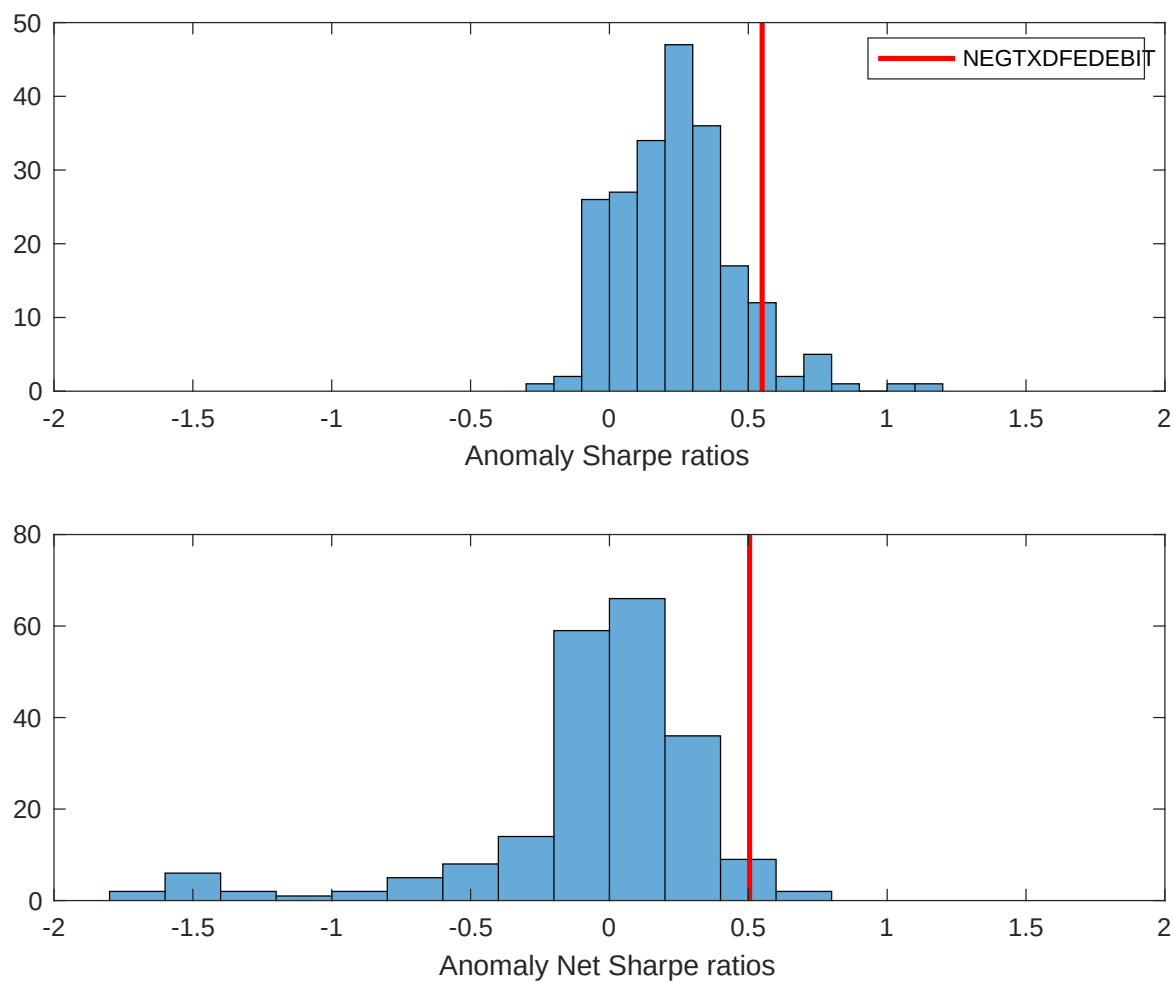


Figure 2: Distribution of Sharpe ratios.

This figure plots a histogram of Sharpe ratios for 212 anomalies, and compares the Sharpe ratio of the TDB with them (red vertical line). Panel A plots results for gross Sharpe ratios. Panel B plots results for net Sharpe ratios.

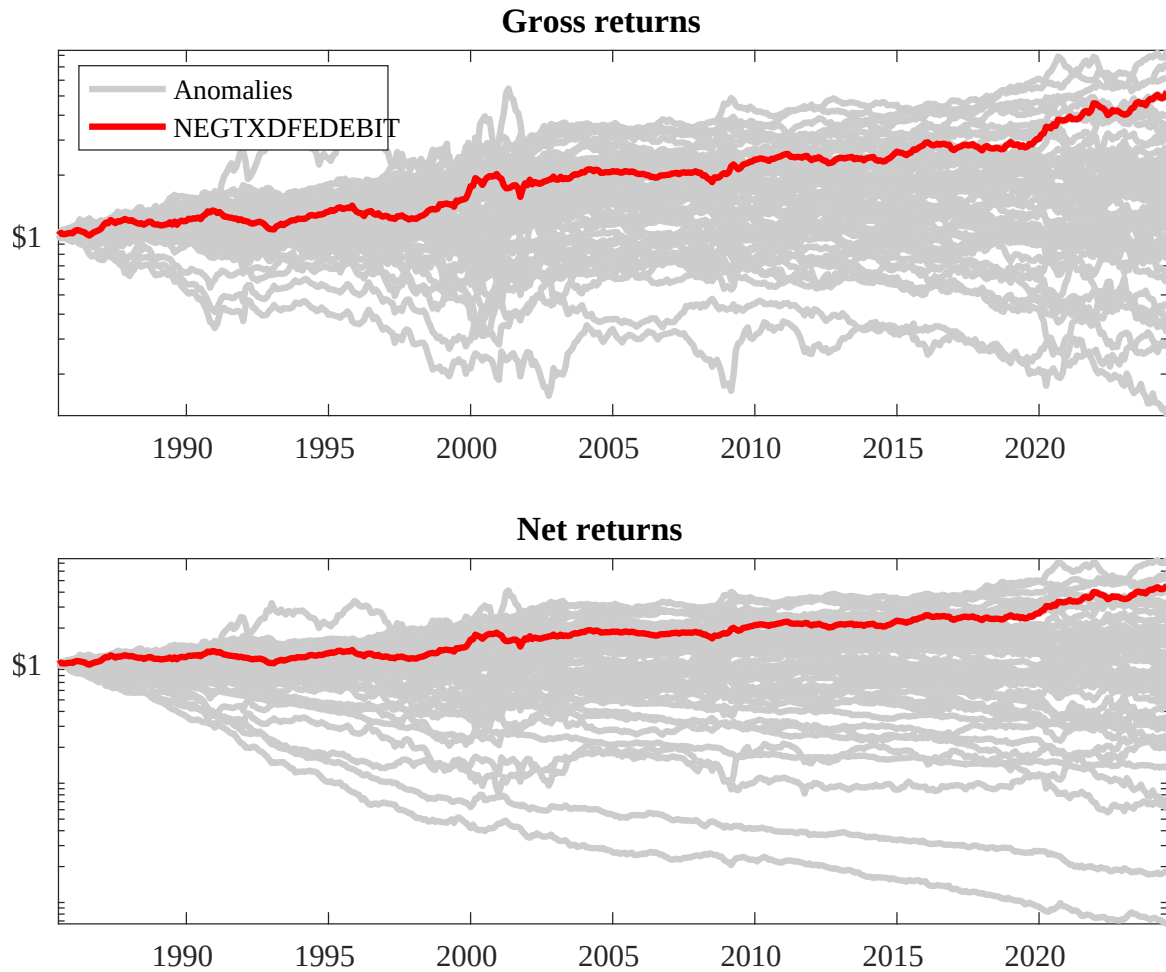
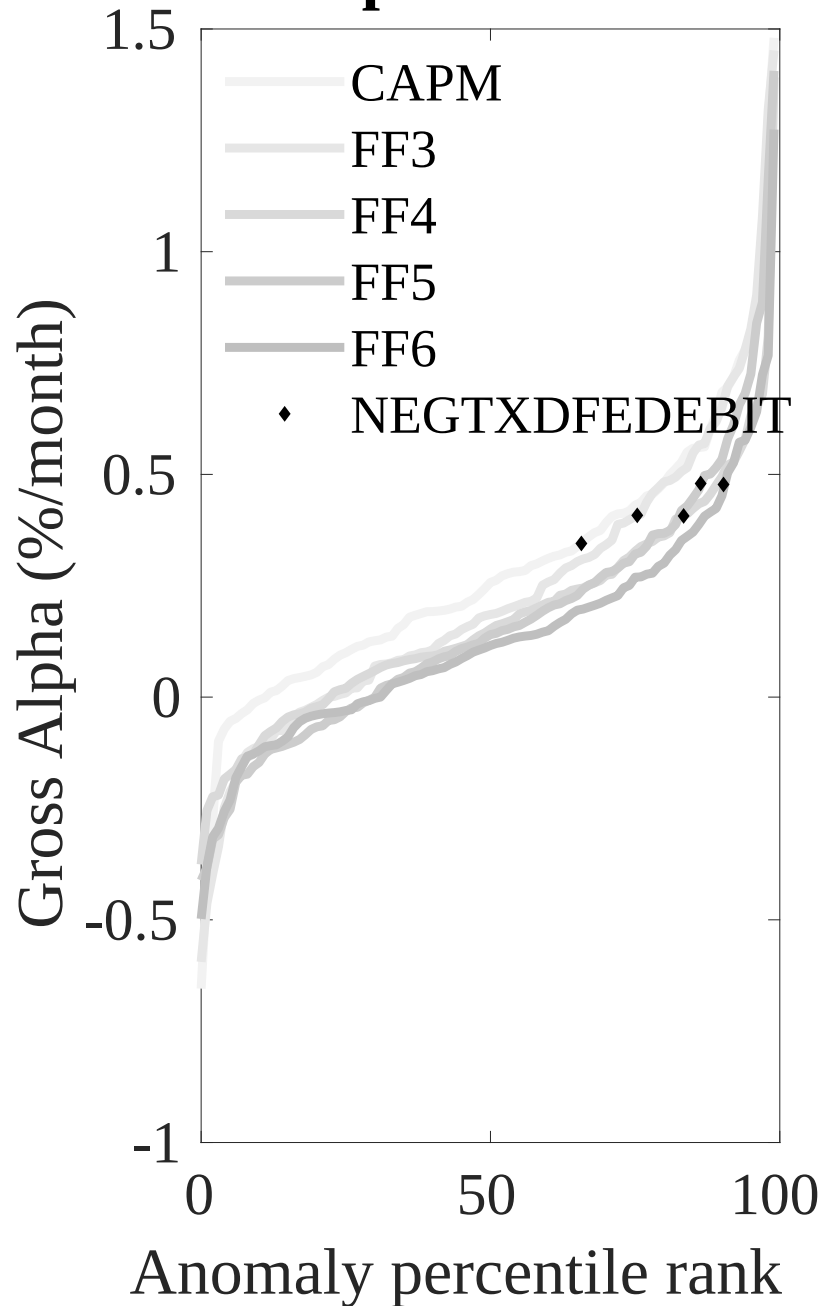


Figure 3: Dollar invested.

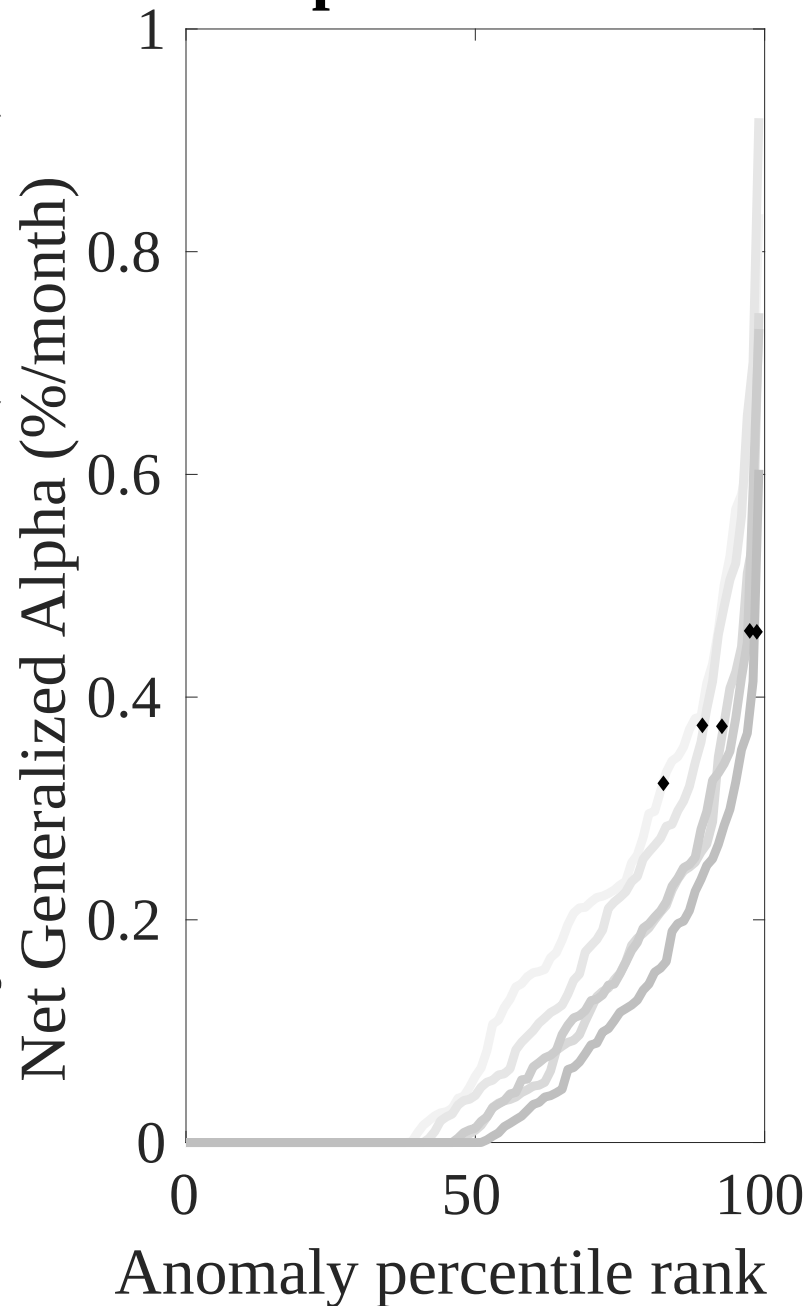
This figure plots the growth of a \$1 invested in 212 anomaly trading strategies (gray lines), and compares those with the TDB trading strategy (red line). The strategies are constructed using value-weighted quintile sorts using NYSE breakpoints. Panel A plots results for gross strategy returns. Panel B plots results for net strategy returns.

Gross Alpha Percentiles



Novy-Marx and Velikov (RFS, 2016)

Net Alpha Percentiles



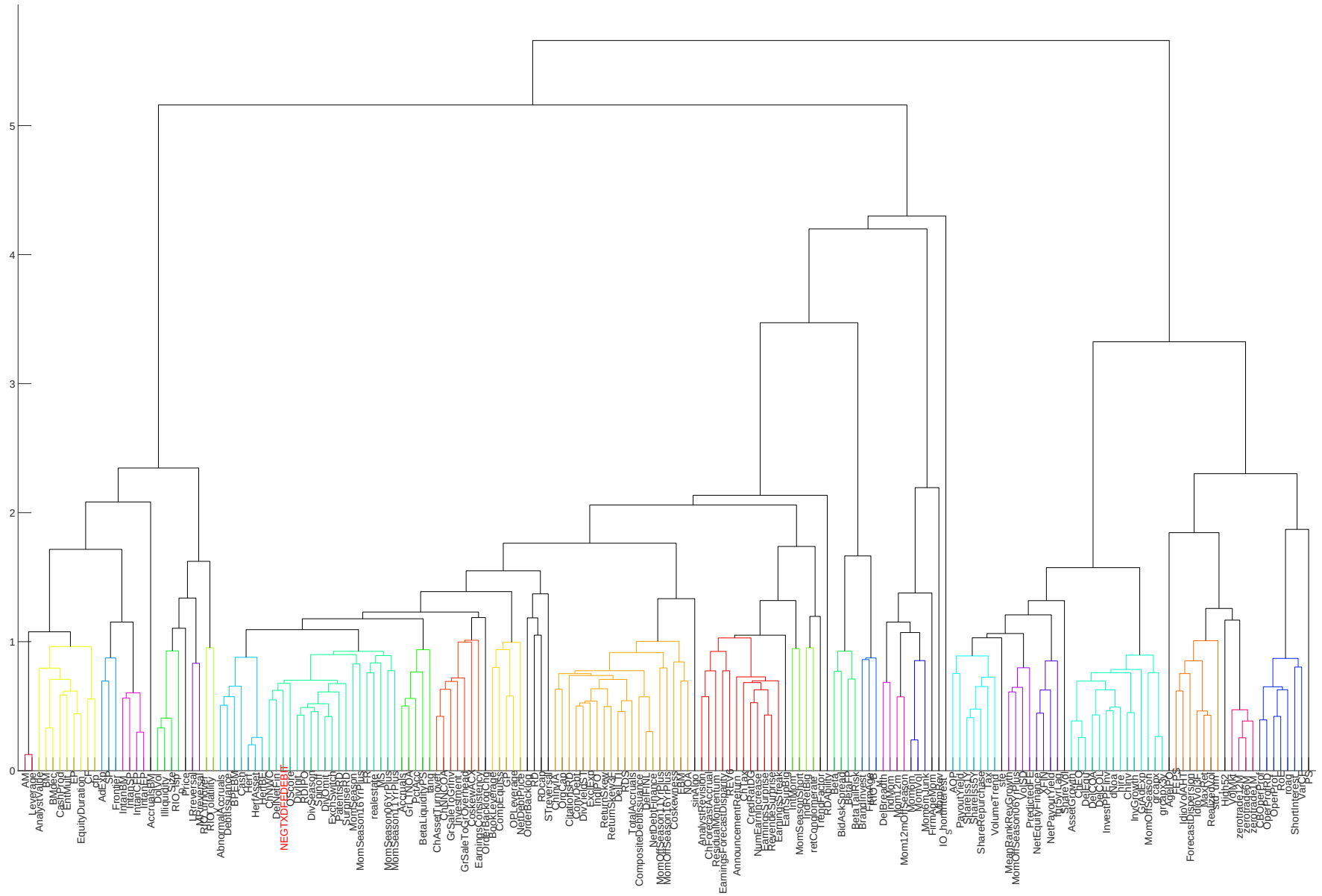


Figure 6: Agglomerative hierarchical cluster plot

This figure plots an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

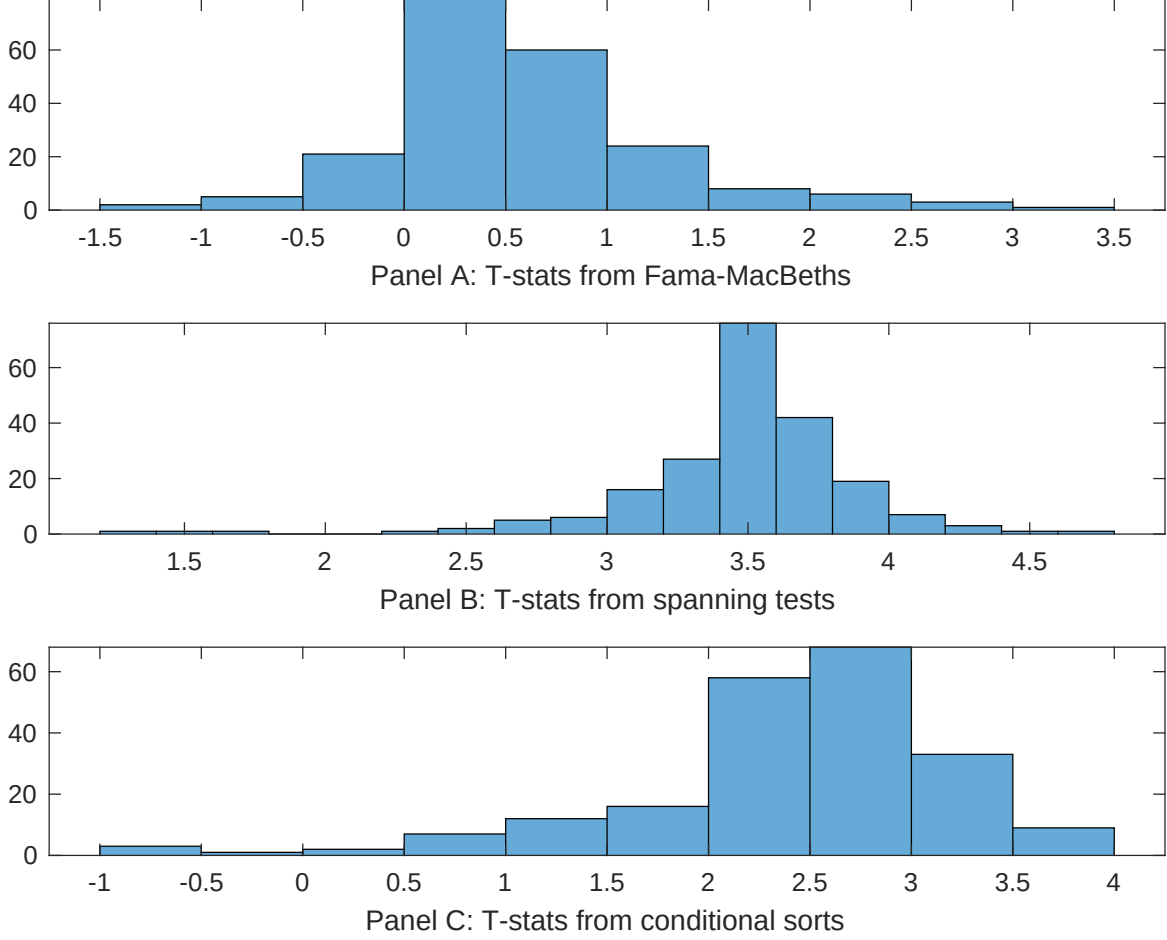


Figure 7: Distribution of t-stats on conditioning strategies

This figure plots histograms of t-statistics for predictability tests of TDB conditioning on each of the 209 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{TDB} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{TDB}TDB_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 209 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{TDB,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 209 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 209 filtered anomaly signals at a time. Then, within each quintile, we sort stocks into quintiles based on TDB. Stocks are finally grouped into five TDB portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted TDB trading strategies conditioned on each of the 209 filtered anomalies.

Table 4: Fama-MacBeths controlling for most closely related anomalies

This table presents Fama-MacBeth results of returns on TDB. and the six most closely related anomalies. The regressions take the following form: $r_{i,t} = \alpha + \beta_{TDB}TDB_{i,t} + \sum_{k=1}^s \beta_{X_k}X_{i,t}^k + \epsilon_{i,t}$. The six most closely related anomalies, X , are Cash to assets, Equity Duration, Cash flow to market, Book to market using December ME, Market leverage, Operating Cash flows to price. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 198506 to 202406.

Intercept	0.10 [3.69]	0.16 [6.32]	0.10 [3.50]	0.93 [3.04]	0.10 [3.51]	0.10 [3.46]	0.12 [4.81]
TDB	0.78 [1.35]	0.18 [0.31]	0.24 [0.43]	0.34 [0.59]	0.39 [0.69]	0.34 [0.59]	0.43 [0.77]
Anomaly 1	0.62 [1.56]						0.11 [3.13]
Anomaly 2		0.25 [2.66]					0.23 [3.41]
Anomaly 3			0.18 [0.58]				-0.23 [-0.98]
Anomaly 4				0.27 [3.22]			0.20 [2.84]
Anomaly 5					0.13 [0.61]		-0.25 [-1.32]
Anomaly 6						0.85 [2.95]	0.79 [4.36]
# months	463	468	463	463	463	463	463
$\bar{R}^2(\%)$	1	0	1	0	0	1	0

Table 5: Spanning tests controlling for most closely related anomalies

This table presents spanning tests results of regressing returns to the TDB trading strategy on trading strategies exploiting the six most closely related anomalies. The regressions take the following form: $r_t^{TDB} = \alpha + \sum_{k=1}^6 \beta_{X_k} r_t^{X_k} + \sum_{j=1}^6 \beta_{f_j} r_t^{f_j} + \epsilon_t$, where X_k indicates each of the six most-closely related anomalies and f_j indicates the six factors from the [Fama and French \(2015\)](#) five-factor model augmented with the [Carhart \(1997\)](#) momentum factor. The six most closely related anomalies, X , are Cash to assets, Equity Duration, Cash flow to market, Book to market using December ME, Market leverage, Operating Cash flows to price. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 198506 to 202406.

Intercept	0.28 [2.84]	0.47 [4.63]	0.48 [4.73]	0.46 [4.56]	0.47 [4.56]	0.51 [5.00]	0.29 [2.90]
Anomaly 1	33.90 [8.54]						30.37 [7.01]
Anomaly 2		-13.35 [-2.49]					-1.87 [-0.29]
Anomaly 3			-8.48 [-2.40]				2.14 [0.48]
Anomaly 4				-25.89 [-3.87]			-9.98 [-1.45]
Anomaly 5					-12.58 [-3.00]		-6.98 [-1.62]
Anomaly 6						-13.55 [-3.62]	-3.78 [-0.87]
mkt	-6.58 [-2.77]	-0.34 [-0.14]	-0.21 [-0.09]	-1.30 [-0.54]	1.38 [0.56]	0.27 [0.11]	-5.33 [-2.14]
smb	-6.87 [-2.00]	-5.10 [-1.33]	-7.16 [-1.95]	-0.88 [-0.22]	-7.09 [-1.94]	-5.87 [-1.60]	-3.35 [-0.86]
hml	-12.23 [-2.75]	-10.22 [-1.36]	-19.96 [-4.06]	1.74 [0.21]	-8.23 [-1.16]	-14.92 [-2.86]	9.45 [1.00]
rmw	-5.99 [-1.26]	-22.11 [-4.77]	-19.30 [-4.01]	-29.19 [-5.93]	-24.13 [-5.18]	-17.73 [-3.71]	-10.77 [-1.97]
cma	18.37 [2.92]	1.37 [0.20]	8.59 [1.29]	6.95 [1.07]	3.20 [0.49]	8.92 [1.36]	15.48 [2.28]
umd	0.42 [0.20]	0.48 [0.21]	-2.98 [-1.07]	1.58 [0.69]	-2.06 [-0.83]	-4.75 [-1.73]	-1.50 [-0.49]
# months	464	468	464	464	464	464	464
$\bar{R}^2(\%)$	29	19	18	20	19	19	29

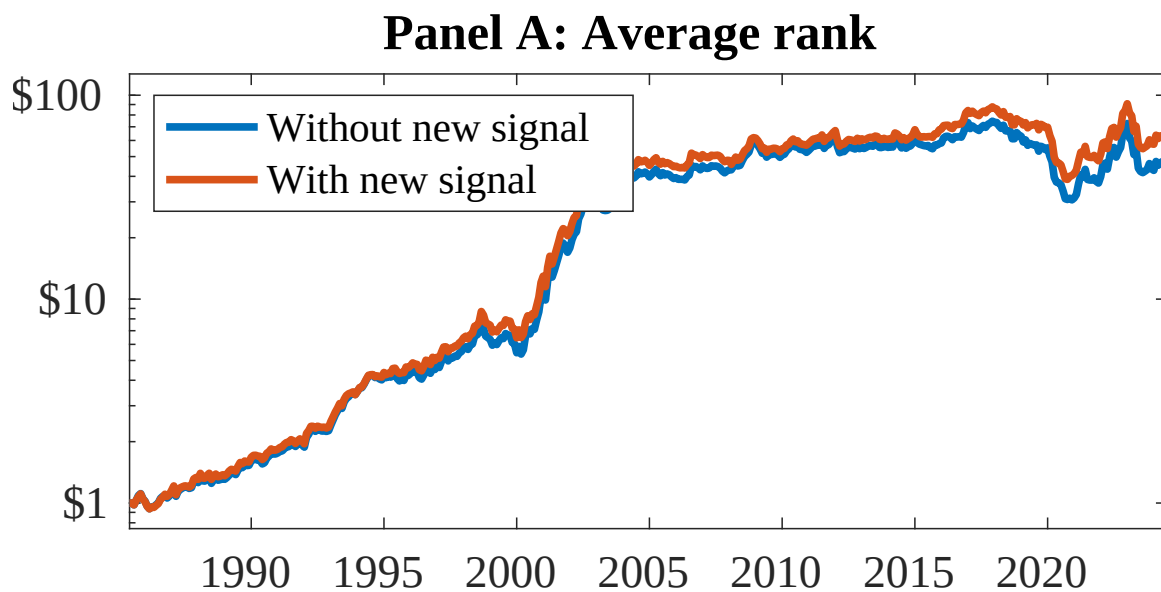


Figure 8: Combination strategy performance

This figure plots the growth of a \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). In all panels, the blue solid lines indicate combination trading strategies that utilize 158 anomalies. The red solid lines indicate combination trading strategies that utilize the 158 anomalies as well as TDB. Panel A shows results using "Average rank" as the combination method. See [Section 7](#) for details on the combination methods.

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